

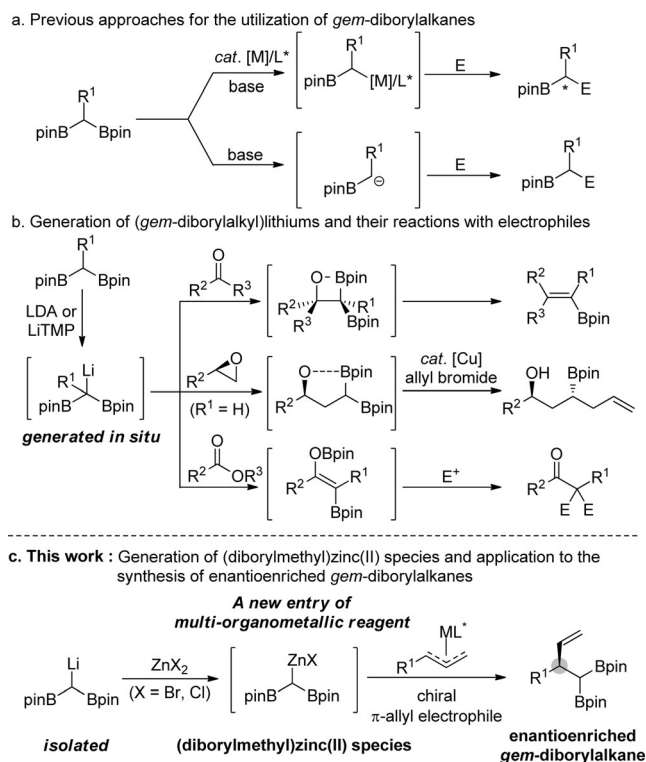
Organometallics

International Edition: DOI: 10.1002/anie.201805476
German Edition: DOI: 10.1002/ange.201805476Generation and Application of (Diborylmethyl)zinc(II) Species: Access to Enantioenriched *gem*-Diborylalkanes by an Asymmetric Allylic Substitution

Yeosan Lee, Jinyoung Park, and Seung Hwan Cho*

Abstract: We report the successful generation of (diborylmethyl)zinc(II) species by transmetalation between isolable (diborylmethyl)lithium and zinc(II) halide ($X = \text{Br}, \text{Cl}$) and their application in the synthesis of enantioenriched *gem*-diborylalkanes bearing a stereogenic center at the β -position of the diboryl groups by an asymmetric allylic substitution reaction. The reaction has a broad substrate scope, and various enantioenriched *gem*-diborylalkanes can be obtained in good yields with excellent enantioselectivity. Further elaboration of the enantioenriched *gem*-diborylalkanes provides access to a diverse set of valuable chiral building blocks.

The preparation of novel multi-organometallics is one of the most important tasks in modern chemistry because they provide access to synthetically valuable compounds by chemoselective carbon-carbon bond-forming reactions.^[1] In particular, the preparation of new types of *gem*-diborylalkanes has gained considerable attention in recent years^[2] owing to the widespread use of these compounds in several transition-metal-catalyzed^[3–5] and transition-metal-free^[6] transformations (Scheme 1 a). Alternatively, *gem*-diborylalkanes have also been used to generate (*gem*-diborylalkyl)lithiums by treatment with a sterically hindered amide base (Scheme 1 b).^[7] These lithium species were known to undergo Boron-Wittig reaction with carbonyl compounds via four-membered cyclic boronate intermediates.^[8] Meek et al. disclosed a highly stereoselective addition of generated (diborylmethyl)lithium to enantioenriched epoxides to obtain five-membered cyclic boronates and their practical use in the copper-catalyzed stereospecific deborylative coupling with allylic electrophiles.^[9] More recently, the group of Pattison reported a convenient method to generate boron enolates by reacting esters with (*gem*-diborylalkyl)lithiums, which could be trapped by an electrophile to afford α,α -difunctionalized ketones.^[10] Despite this progress, the current utilization of (*gem*-diborylalkyl)metallics relies primarily on the use of in situ generated (*gem*-diborylalkyl)lithiums. Such reagents are useful for reactions with organo electrophiles, but there have been problems in developing catalytic asymmetric reactions

Scheme 1. Use of *gem*-diborylalkanes. pin = pinacol.

due to their high reactivity. Therefore, the development of an efficient and operationally simple protocol for the preparation of other (*gem*-diborylalkyl)metallics and application of these species to catalytic asymmetric reactions is highly desirable.

Herein, we report a convenient approach to generate (*gem*-diborylalkyl)zinc(II) species, comprising two identical boron groups and zinc species at the same sp^3 carbon center, by transmetalation of isolated (diborylmethyl)lithium with ZnX_2 ($X = \text{Br}, \text{Cl}$). This newly prepared organometallic reagent was applied in metal-catalyzed asymmetric allylic substitution reaction (Scheme 1 c)^[11] to afford a broad range of enantioenriched *gem*-diborylalkanes in high yields with excellent enantioselectivity. The obtained enantioenriched *gem*-diborylalkanes can be elaborated to various synthetically valuable chiral building blocks via appropriate stereospecific transformations, highlighting the synthetic utility of these new compounds.

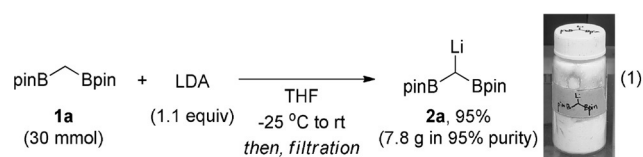
Transition-metal-catalyzed asymmetric allylic alkylation (AAA) reactions are among the most efficient and reliable tools to prepare enantioenriched allylic compounds, which

[*] Y. Lee, J. Park, Prof. Dr. S. H. Cho
Department of Chemistry
Pohang University of Science and Technology (POSTECH)
Pohang, 37673 (Rep. of Korea)
E-mail: seunghwan@postech.ac.kr

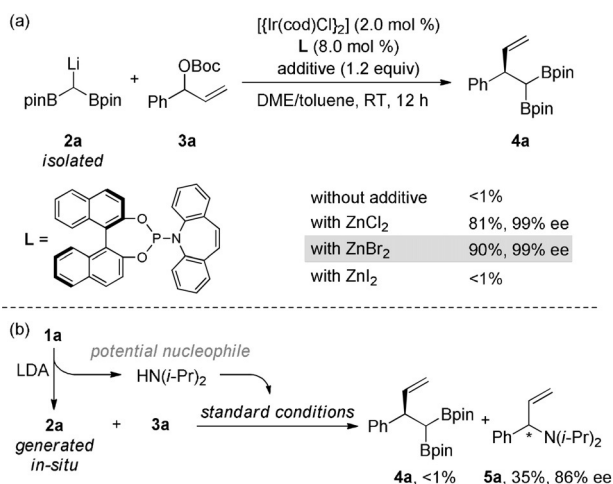
Supporting information and the ORCID identification number(s) for the author(s) of this article can be found under:
<https://doi.org/10.1002/anie.201805476>.

are useful building blocks in organic synthesis.^[12] Recent progress in this field includes the development of efficient protocols for the introduction of a range of nucleophiles to the branched position of allylic electrophiles with excellent enantioselectivity.^[13,14] Based on these substantial advances, we assumed that an iridium-catalyzed AAA reaction with (*gem*-diborylalkyl)metallics^[14] would offer a convenient platform for a rapid accessing of enantioenriched *gem*-diborylalkanes.

Before commencing our study, we first attempted to establish a method for the isolation of (diborylmethyl)lithium (**2a**). In previous reports, **2a** has been typically generated *in situ* from diborylmethane (**1a**) in the presence of a sterically hindered amide base, such as lithium diisopropylamide (LDA) or lithium tetramethylpiperidide (LiTMP); however, this approach was accompanied by the formation of an amine after deprotonation of the α -C–H bond of **1a**, which might lead to the formation of an undesired allylic amination product during the iridium-catalyzed AAA process.^[15] Therefore, we deemed the isolation of **2a** was essential to efficiently achieve our proposed transformation while avoiding the problematic, competing allylic amination pathway. To the best of our knowledge, **2a** has never been isolated and indeed, we successfully isolated **2a** in 95% yield with 95% purity^[16] as a white solid from the reaction of **1a** (30 mmol) with LDA in THF and subsequent filtration inside a glove box [Eq. (1)].^[17]



To examine the feasibility of our proposed reaction, we initially employed **2a** as a potential nucleophile for the iridium-catalyzed AAA reaction (Scheme 2a). No reaction occurred when Boc-protected racemic allylic alcohol (**3a**) and

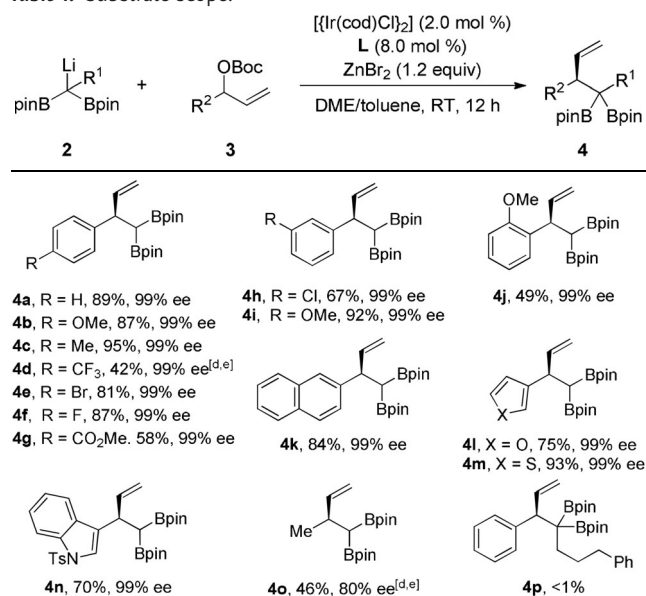


Scheme 2. a) *In situ* generation of (diborylmethyl)zinc(II) species from isolable (diborylmethyl)lithium and application to the synthesis of enantioenriched *gem*-diborylalkane through iridium-catalyzed AAA. b) Reaction of **3a** with *in situ* generated **2a**.

2a were reacted in the presence of 2.0 mol % [[Ir(cod)Cl]₂] and 8.0 mol % phosphoramidite ligand (**L**), one of the most frequently used ligands for the asymmetric allylic substitution of branched allylic alcohols or carbonates,^[18] in DME/toluene at room temperature, and all of the starting materials were decomposed after 12 h. However, the addition of ZnCl₂, as an additive, did furnish the enantioenriched homoallylic *gem*-diboronate **4a** in 81% yield and 99% ee. This result indicated that the (diborylmethyl)zinc(II) species^[19] generated *in situ* served as a nucleophile under the reaction conditions.^[20,21] A higher yield of 90% was obtained when ZnBr₂ was employed instead of ZnCl₂, but the addition of ZnI₂ deteriorated the reactivity.^[17] Note that the reaction of **3a** with **2a** generated *in situ* from diborylmethane and LDA in the presence of ZnBr₂ did not afford the enantioenriched product **4a**, instead, provided allylic amine **5a** as a single product in 35% yield and 86% ee (Scheme 2b). This result clearly suggests that the use of isolated **2a** is crucial to achieve the present transformation. The absolute configuration of **4a** was assigned to be *R* by further derivatization of **4a** to homoallylic boronate ester **6** (vide infra) and comparison of its optical rotation with that in the literature.^[5f,g,17]

Having achieved this result, the scope of allylic electrophiles was immediately investigated (Table 1). The reactions of allylic electrophiles bearing electron-donating (**4b** and **4c**) and electron-withdrawing (**4d**) substituents at the C4-position on the arene ring provided the corresponding enantioenriched *gem*-diborylalkanes in moderate to good yields with excellent enantioselectivity, although **4d** required increased catalyst loading and elevated temperature (50 °C) to give a synthetically acceptable yield. The reaction condition was

Table 1: Substrate scope.^[a–c]



[a] Reaction conditions: **2** (1.1 equiv), **3** (0.20 mmol), ZnBr₂ (1.2 equiv), [[Ir(cod)Cl]₂] (2.0 mol %), **L** (8.0 mol %), and DME/toluene (1.0 mL, v/v = 1:1) at room temperature for 12 h. [b] Isolated yields are given. [c] Enantiomeric excess was determined by HPLC. [d] 4.0 mol % of [[Ir(cod)Cl]₂] and 16 mol % **L** were used. [e] Runs at 50 °C. Ts = *p*-toluenesulfonyl.

compatible with substrates bearing halides (**4e** and **4f**) or an ester (**4g**). Racemic allylic electrophiles containing substituents at the C3- or C2- position on the arene ring afforded the products **4h** and **4i** with high enantioselectivity, although the use of an allylic electrophile bearing a substituent at the C2-position of the arene ring (**4j**) led to decreased yield. The reaction conducted with an allylic carbonate containing a naphthyl group furnished **4k** in good yield and enantioselectivity. The analogous reactions with heteroaryl allylic carbonates containing 3-furanyl, 3-thienyl, and 3-indolyl groups yielded enantioenriched *gem*-diborylalkanes **4l–4n** with good to excellent efficiency and enantioselectivities. In the presence of 4.0 mol % $[\text{Ir}(\text{COD})\text{Cl}]_2$ and 16 mol % **L** at 50°C, an aliphatic allylic carbonates also underwent the reaction and provided the product **4o** in moderate yield and enantioselectivity. When (*gem*-diborylpropyl-3-phenyl)-lithium (**2b**) was subjected to the optimized reaction conditions, no desired product (**4p**) was obtained, indicating the current limitation of the developed process.

A gram scale reaction was conducted to demonstrate the practicality of the current methodology (Scheme 3a). When 5.0 mmol of **3a** and **2a** were subjected to the reaction conditions, the product **4a** was obtained in 82% yield (1.57 g) without compromising the enantioselectivity. Given the broad applicability of the *gem*-diborylalkanes and usefulness of the enantioenriched allylic compounds in synthetic chemistry, we elaborated **4a** to various enantioenriched compounds (Scheme 3b). The protodeboronation of **4a** by treatment with NaOtBu and MeOH in THF at room temperature afforded homoallylic boronate ester **6** in 82% yield and

99% *ee*. The base-promoted, direct C2-selective alkylation of **3a** with quinoline *N*-oxide formed **7** in 71% yield without affecting the enantioselectivity upon treatment with NaOMe in toluene at 120°C.^[6c,22] The hydrogenation of an olefin moiety of **4a** using Pd/C in MeOH at room temperature, followed by the oxidation of diboron groups in the presence of H_2O_2 and NaHCO_3 in $\text{THF}/\text{H}_2\text{O}$ furnished the corresponding aldehyde **8** in 67% yield (2 steps) in 98% *ee*. Treatment of **4a** with LDA in THF and subsequent trapping of the generated (*gem*-diborylalkyl)lithium intermediate with 3-phenylpropylbromide yielded internal *gem*-diborylalkane **4p** in 94% yield, respectively, without erosion of the enantiomeric excess (99% *ee*). In view of the poor reactivity of (*gem*-diborylalkyl)zinc(II) bromide in the iridium-catalyzed AAA reaction (Table 1, **4p**), this approach provided an alternative route for the synthesis of enantioenriched internal *gem*-diborylalkanes. Following olefin hydrogenation of **4a**, the reaction of (*gem*-diborylalkyl)lithium with diiodomethane afforded enantioenriched 1,1-disubstituted vinylboronate **9** (75% in 2 steps from **4a**) under slightly modified Morken's conditions.^[8a] The obtained product **9** converted to enantioenriched acyclic α -substituted ketone **10** (97%, 99% *ee*) by Bpin oxidation. Compound **9** was also used in a Suzuki–Miyaura cross-coupling with 4-iodoanisole catalyzed by $\text{Pd}(\text{PPh}_3)_4$ to afford product **11** in 80% and 99% *ee*.

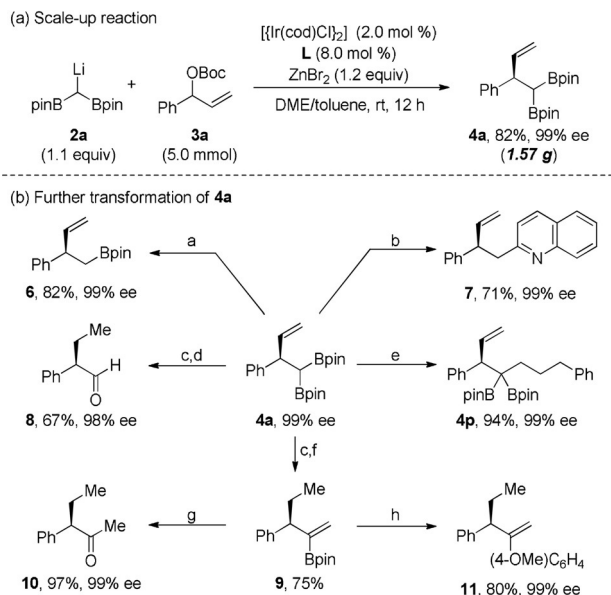
In summary, we have developed an efficient protocol for the generation of (diborylmethyl)zinc(II) species from isolable (diborylmethyl)lithium with zinc(II) halides and demonstrated the successful application of this novel multi-organometallic reagents for the synthesis of enantiomerically enriched *gem*-diborylalkanes by an iridium-catalyzed asymmetric allylic alkylation reaction. The reaction provides a broad range of enantioenriched *gem*-diborylalkanes in good yields and excellent enantioselectivity. Synthetic applications of the obtained products are also demonstrated, and they represent useful enantioenriched compounds in organic synthesis. Further efforts to generate other (*gem*-diborylalkyl)metallic species and develop metal-catalyzed chemoselective and stereoselective transformations are ongoing in our laboratory.

Acknowledgements

This work was supported by the Samsung Science and Technology Foundation under Project Number SSTF-BA1701-11. Y. Lee (NRF-2017H1A2A1045655) and J. Park (NRF-2016H1A2A1908616) thank the National Research Foundation of Korea (NRF) for the global Ph.D. fellowship. S. H. Cho thanks POSCO Green Science Program and Cheongam Science Fellowship for their generous support. We thank Professor Jun Hee Lee (Dongguk University) for helpful comments on the manuscript.

Conflict of interest

The authors declare no conflict of interest.



Scheme 3. Transformations of the enantioenriched **4a**: Reaction conditions: [a] NaOtBu , MeOH , THF, RT, 6 h. [b] Quinoline-*N*-oxide, NaOMe , toluene, 120°C, 3 h. [c] cat. Pd/C , H_2 , MeOH , RT, 2 h. [d] H_2O_2 , NaHCO_3 , $\text{THF}/\text{H}_2\text{O}$, 0°C to RT, 1 h. [e] LDA , 3-phenylpropyl bromide, THF, 0°C to RT, 6 h. [f] LDA , CH_2I_2 , THF, 0°C to 60°C, 48 h. [g] $\text{NaBO}_3 \cdot 4\text{H}_2\text{O}$, $\text{THF}/\text{H}_2\text{O}$, 0°C to RT, 3 h. [h] cat. $\text{Pd}(\text{PPh}_3)_4$, 4-iodoanisole, NaOH , 1,4-dioxane/ H_2O , 100°C, 6 h.

Keywords: allylic alkylation · enantioselectivity · homogeneous catalysis · organoboron compounds · zinc

How to cite: *Angew. Chem. Int. Ed.* **2018**, 57, 12930–12934
Angew. Chem. **2018**, 130, 13112–13116

- [1] a) C. Sandford, V. K. Aggarwal, *Chem. Commun.* **2017**, 53, 5481; b) A. H. Cherney, N. T. Kadunce, S. E. Reisman, *Chem. Rev.* **2015**, 115, 9587; c) L. Xu, S. Zhang, P. Li, *Chem. Soc. Rev.* **2015**, 44, 8848; d) J. P. G. Rygus, C. M. Crudden, *J. Am. Chem. Soc.* **2017**, 139, 18124.
- [2] a) H. Abu Ali, I. Goldberg, D. Kaufmann, C. Burmeister, M. Srebnik, *Organometallics* **2002**, 21, 1870; b) K. Endo, M. Hirokami, T. Shibata, *Synlett* **2009**, 1331; c) S. H. Cho, J. F. Hartwig, *Chem. Sci.* **2014**, 5, 694; d) S. Lee, D. Li, J. Yun, *Chem. Asian J.* **2014**, 9, 2440; e) H. Li, X. Shangguan, Z. Zhang, S. Huang, Y. Zhang, J. Wang, *Org. Lett.* **2014**, 16, 448; f) M. L. Scheuermann, E. J. Johnson, P. J. Chirik, *Org. Lett.* **2015**, 17, 2716; g) W. N. Palmer, J. V. Obligacion, I. Pappas, P. J. Chirik, *J. Am. Chem. Soc.* **2016**, 138, 766; h) G. Gao, J. Yan, K. Yang, F. Chen, Q. Song, *Green Chem.* **2017**, 19, 3997; i) L. Li, T. Gong, X. Lu, B. Xiao, Y. Fu, *Nat. Commun.* **2017**, 8, 345; j) L. Wang, T. Zhang, W. Sun, Z. He, C. Xia, Y. Lan, C. Liu, *J. Am. Chem. Soc.* **2017**, 139, 5257; k) H. Zhao, M. Tong, H. Wang, S. Xu, *Org. Biomol. Chem.* **2017**, 15, 3418; l) W. J. Teo, S. Ge, *Angew. Chem. Int. Ed.* **2018**, 57, 1654; *Angew. Chem.* **2018**, 130, 1670.
- [3] a) N. Miralles, R. J. Maza, E. Fernández, *Adv. Synth. Catal.* **2018**, 360, 1306; b) R. Nallagonda, K. Padala, A. Masarwa, *Org. Biomol. Chem.* **2018**, 16, 1050; c) C. Wu, J. Wang, *Tetrahedron Lett.* **2018**, 59, 2128.
- [4] Examples for catalytic non-enantioselective reactions, see: a) K. Endo, T. Ohkubo, M. Hirokami, T. Shibata, *J. Am. Chem. Soc.* **2010**, 132, 11033; b) H. Li, Z. Zhang, X. Shangguan, S. Huang, J. Chen, Y. Zhang, J. Wang, *Angew. Chem. Int. Ed.* **2014**, 53, 11921; *Angew. Chem.* **2014**, 126, 12115; c) M. V. Joannou, B. S. Moyer, M. J. Goldfogel, S. J. Meek, *Angew. Chem. Int. Ed.* **2015**, 54, 14141; *Angew. Chem.* **2015**, 127, 14347; d) L.-C. Cui, Z.-Q. Zhang, X. Lu, B. Xiao, Y. Fu, *RSC Adv.* **2016**, 6, 51932; e) A. Ebrahim-Alkhalil, Z.-Q. Zhang, T.-J. Gong, W. Su, X.-Y. Lu, B. Xiao, Y. Fu, *Chem. Commun.* **2016**, 52, 4891; f) J. Kim, S. Park, J. Park, S. H. Cho, *Angew. Chem. Int. Ed.* **2016**, 55, 1498; *Angew. Chem.* **2016**, 128, 1520; g) Z.-Q. Zhang, B. Zhang, X. Lu, J.-H. Liu, X.-Y. Lu, B. Xiao, Y. Fu, *Org. Lett.* **2016**, 18, 952; h) F. Li, Z.-Q. Zhang, X. Lu, B. Xiao, Y. Fu, *Chem. Commun.* **2017**, 53, 3551.
- [5] Examples for catalytic enantioselective reactions, see: a) B. Potter, A. A. Szymaniak, E. K. Edelstein, J. P. Morken, *J. Am. Chem. Soc.* **2014**, 136, 17918; b) C. Sun, B. Potter, J. P. Morken, *J. Am. Chem. Soc.* **2014**, 136, 6534; c) M. V. Joannou, B. S. Moyer, S. J. Meek, *J. Am. Chem. Soc.* **2015**, 137, 6176; d) S. A. Murray, J. C. Green, S. B. Taylor, S. J. Meek, *Angew. Chem. Int. Ed.* **2016**, 55, 9065; *Angew. Chem.* **2016**, 128, 9211; e) J. Park, Y. Lee, J. Kim, S. H. Cho, *Org. Lett.* **2016**, 18, 1210; f) Y. Shi, A. H. Hoveyda, *Angew. Chem. Int. Ed.* **2016**, 55, 3455; *Angew. Chem.* **2016**, 128, 3516; g) M. Zhan, R.-Z. Li, Z.-D. Mou, C.-G. Cao, J. Liu, Y.-W. Chen, D. Niu, *ACS Catal.* **2016**, 6, 3381; h) J. Kim, K. Ko, S. H. Cho, *Angew. Chem. Int. Ed.* **2017**, 56, 11584; *Angew. Chem.* **2017**, 129, 11742; i) T. Miura, J. Nakahashi, M. Murakami, *Angew. Chem. Int. Ed.* **2017**, 56, 6989; *Angew. Chem.* **2017**, 129, 7093; j) T. Miura, J. Nakahashi, W. Zhou, Y. Shiratori, S. G. Stewart, M. Murakami, *J. Am. Chem. Soc.* **2017**, 139, 10903.
- [6] a) K. Hong, X. Liu, J. P. Morken, *J. Am. Chem. Soc.* **2014**, 136, 10581; b) J. R. Coombs, L. Zhang, J. P. Morken, *J. Am. Chem. Soc.* **2014**, 136, 16140; c) W. Jo, J. Kim, S. Choi, S. H. Cho, *Angew. Chem. Int. Ed.* **2016**, 55, 9690; *Angew. Chem.* **2016**, 128, 9842; d) J. Kim, S. H. Cho, *Synlett* **2016**, 27, 2525; e) D. J. Blair, D. Tanini, J. M. Bateman, H. K. Scott, E. L. Myers, V. K. Aggarwal, *Chem. Sci.* **2017**, 8, 2898; f) C. Hwang, W. Jo, S. H. Cho, *Chem. Commun.* **2017**, 53, 7573; g) Y. Lee, S.-y. Baek, J. Park, S.-T. Kim, S. Tussupbayev, J. Kim, M.-H. Baik, S. H. Cho, *J. Am. Chem. Soc.* **2017**, 139, 976; h) J. Park, S. Choi, Y. Lee, S. H. Cho, *Org. Lett.* **2017**, 19, 4054; i) A. Yamazaki, K. Nagao, T. Iwai, H. Ohmiya, M. Sawamura, *Angew. Chem. Int. Ed.* **2018**, 57, 3196; *Angew. Chem.* **2018**, 130, 3250; j) W. Sun, L. Wang, C. Xia, C. Liu, *Angew. Chem. Int. Ed.* **2018**, 57, 5501; *Angew. Chem.* **2018**, 130, 5599.
- [7] a) D. S. Matteson, P. K. Jesthi, *J. Organomet. Chem.* **1976**, 110, 25; b) D. S. Matteson, R. J. Moody, *J. Am. Chem. Soc.* **1977**, 99, 3196; c) D. S. Matteson, R. J. Moody, *Organometallics* **1982**, 1, 20.
- [8] a) J. R. Coombs, L. Zhang, J. P. Morken, *Org. Lett.* **2015**, 17, 1708; b) T. C. Stephens, G. Pattison, *Org. Lett.* **2017**, 19, 3498.
- [9] a) S. A. Murray, M. Z. Liang, S. J. Meek, *J. Am. Chem. Soc.* **2017**, 139, 14061; b) S. A. Murray, E. C. M. Luc, S. J. Meek, *Org. Lett.* **2018**, 20, 469.
- [10] C. E. Iacono, T. C. Stephens, T. S. Rajan, G. Pattison, *J. Am. Chem. Soc.* **2018**, 140, 2036.
- [11] a) Y. Miguel, O. Rosa, *Eur. J. Org. Chem.* **2004**, 3833; b) *The Chemistry of Organozinc Compounds* (Eds.: Z. Rappoport, I. Marek), Wiley, Chichester, **2006**; c) P. Knochel, R. D. Singer, *Chem. Rev.* **1993**, 93, 2117; d) J. E. Fleckenstein, K. Koszinowski, *Organometallics* **2011**, 30, 5018; e) S. H. Kyne, J. M. Percy, R. D. Pullin, J. M. Redmond, P. G. Wilson, *Org. Biomol. Chem.* **2011**, 9, 8328; f) C. J. Cordier, R. J. Lundgren, G. C. Fu, *J. Am. Chem. Soc.* **2013**, 135, 10946; g) T. Klatt, J. T. Markiewicz, C. Sämman, P. Knochel, *J. Org. Chem.* **2014**, 79, 4253.
- [12] For representative reviews, see: a) B. M. Trost, D. L. Van Vranken, *Chem. Rev.* **1996**, 96, 395; b) B. M. Trost, M. L. Crawley, *Chem. Rev.* **2003**, 103, 2921; c) L.-X. Dai, T. Tu, S.-L. You, W.-P. Deng, X.-L. Hou, *Acc. Chem. Res.* **2003**, 36, 659; d) Z. Lu, S. Ma, *Angew. Chem. Int. Ed.* **2008**, 47, 258; *Angew. Chem.* **2008**, 120, 264.
- [13] For selected reviews, see: a) A. Alexakis, J. E. Backvall, N. Krause, O. Pàmies, M. Diéguez, *Chem. Rev.* **2008**, 108, 2796; b) S. R. Harutyunyan, T. den Hartog, K. Geurts, A. J. Minnaard, B. L. Feringa, *Chem. Rev.* **2008**, 108, 2824; c) J. F. Teichert, B. L. Feringa, *Angew. Chem. Int. Ed.* **2010**, 49, 2486; *Angew. Chem.* **2010**, 122, 2538.
- [14] For reviews of iridium-catalyzed allylic substitutions, see: a) H. Miyabe, Y. Takemoto, *Synlett* **2005**, 1641; b) R. Takeuchi, S. Kozuka, *Synthesis* **2006**, 3349; c) G. Helmchen, A. Dahnz, P. Dubon, M. Schelwies, R. Weihofen, *Chem. Commun.* **2007**, 675; d) J. F. Hartwig, L. M. Stanley, *Acc. Chem. Res.* **2010**, 43, 1461; e) J. F. Hartwig, M. J. Pouy, *Top. Organomet. Chem.* **2011**, 34, 169; f) W.-B. Liu, J.-B. Xia, S.-L. You, *Top. Organomet. Chem.* **2012**, 38, 155; g) P. Tosatti, A. Nelson, S. P. Marsden, *Org. Biomol. Chem.* **2012**, 10, 3147; h) J. C. Hethcox, S. E. Shockley, B. M. Stoltz, *ACS Catal.* **2016**, 6, 6207.
- [15] a) R. Takeuchi, N. Ue, K. Tanabe, K. Yamashita, N. Shiga, *J. Am. Chem. Soc.* **2001**, 123, 9525; b) T. Ohmura, J. F. Hartwig, *J. Am. Chem. Soc.* **2002**, 124, 15164; c) M. Lafrance, M. Roggen, E. M. Carreira, *Angew. Chem. Int. Ed.* **2012**, 51, 3470; *Angew. Chem.* **2012**, 124, 3527; d) W. K. Walker, D. L. Anderson, R. W. Stokes, S. J. Smith, D. J. Michaelis, *Org. Lett.* **2015**, 17, 752; e) X. Zhang, W.-B. Liu, Q. Cheng, S.-L. You, *Organometallics* **2016**, 35, 2467.
- [16] The purity of **2a** was determined by titration using diphenylacetic acid as a titrant. For details, see the Supporting Information.
- [17] See the Supporting Information for details.
- [18] For a recent review, see: a) J. Qu, G. Helmchen, *Acc. Chem. Res.* **2017**, 50, 2539, and references therein. For recent selected examples, see: b) S. L. Rössler, S. Krautwald, E. M. Carreira, *J. Am. Chem. Soc.* **2017**, 139, 3603; c) J. Y. Hamilton, S. L. Rössler, E. M. Carreira, *J. Am. Chem. Soc.* **2017**, 139, 8082; d) T. Sandmeier, S. Krautwald, E. M. Carreira, *Angew. Chem. Int. Ed.* **2017**, 56, 11515; *Angew. Chem.* **2017**, 129, 11673; e) M. A.

- Schafroth, S. M. Rummelt, D. Sarlah, E. M. Carreira, *Org. Lett.* **2017**, *19*, 3235.
- [19] We assumed that the coordination of zinc to one of oxygens of Bpin group might have an effect to stabilize (diborylmethyl)zinc species. For details, see: a) D. S. Matteson, E. Erdik, *Organometallics* **1983**, *2*, 1083; b) D. S. Matteson, K. M. Sadhu, *J. Am. Chem. Soc.* **1983**, *105*, 2077; c) D. S. Matteson, K. M. Sadhu, *Organometallics* **1984**, *3*, 614; d) D. S. Matteson, K. M. Sadhu, M. L. Peterson, *J. Am. Chem. Soc.* **1986**, *108*, 810; e) M. M. Midland, *J. Org. Chem.* **1998**, *63*, 914; f) D. S. Matteson, *J. Org. Chem.* **2013**, *78*, 10009. Further experimental and computational studies are ongoing to elucidate the structure of (diborylmethyl)zinc species.
- [20] a) F. Dübner, P. Knochel, *Angew. Chem. Int. Ed.* **1999**, *38*, 379; *Angew. Chem.* **1999**, *111*, 391; b) C. A. Luchaco-Cullis, H. Mizutani, K. E. Murphy, A. H. Hoveyda, *Angew. Chem. Int. Ed.* **2001**, *40*, 1456; *Angew. Chem.* **2001**, *113*, 1504; c) H. Malda, A. W. van Zijl, L. A. Arnold, B. L. Feringa, *Org. Lett.* **2001**, *3*, 1169; d) K. E. Murphy, A. H. Hoveyda, *J. Am. Chem. Soc.* **2003**, *125*, 4690; e) U. Piarulli, P. Daubos, C. Claverie, M. Roux, C. Gennari, *Angew. Chem. Int. Ed.* **2003**, *42*, 234; *Angew. Chem.* **2003**, *115*, 244; f) A. O. Larsen, W. Leu, C. N. Oberhuber, J. E. Campbell, A. H. Hoveyda, *J. Am. Chem. Soc.* **2004**, *126*, 11130; g) A. W. van Zijl, L. A. Arnold, A. J. Minnaard, B. L. Feringa, *Adv. Synth. Catal.* **2004**, *346*, 413; h) P. J. Goldsmith, S. J. Teat, S. Woodward, *Angew. Chem. Int. Ed.* **2005**, *44*, 2235; *Angew. Chem.* **2005**, *117*, 2275; i) N. Yoshikai, K. Miura, E. Nakamura, *Adv. Synth. Catal.* **2009**, *351*, 1014; j) A. Misale, S. Niyomchon, M. Luparia, N. Maulide, *Angew. Chem. Int. Ed.* **2014**, *53*, 7068; *Angew. Chem.* **2014**, *126*, 7188; k) J. H. Kim, Y. O. Ko, J. Bouffard, S.-g. Lee, *Chem. Soc. Rev.* **2015**, *44*, 2489.
- [21] a) D. Polet, X. Rathgeb, C. A. Falciola, J.-B. Langlois, S. E. Hajjaji, A. Alexakis, L. Jean-Baptiste, E. H. Samir, A. Alexandre, *Chem. Eur. J.* **2009**, *15*, 1205; b) J. Y. Hamilton, D. Sarlah, E. M. Carreira, *Angew. Chem. Int. Ed.* **2015**, *54*, 7644; *Angew. Chem.* **2015**, *127*, 7754.
- [22] X.-J. Liu, S.-L. You, *Angew. Chem. Int. Ed.* **2017**, *56*, 4002; *Angew. Chem.* **2017**, *129*, 4060.

Manuscript received: May 11, 2018

Accepted manuscript online: July 2, 2018

Version of record online: July 26, 2018